Yoshwan Pathipati

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Education

Virginia Tech Blacksburg, VA
B.S. in Computer Science; Minors in Artificial Intelligence, Cybersecurity, and Mathematics *Aug. 2023 – May 2027*

- **Honors:** Dean's List (top 20%) **Leadership:** Co-founder, Collaborative Coding Activities Club (80+ members)
- **Coursework:** Machine Learning, Data Structures & Algorithms, Cloud Software Development, Cryptography

Experience

AI Research Assistant, LLM Automation Sep. 2026 – Present
Virginia Tech, Dept. of Engineering Education *Blacksburg, VA*

- Develop Python preprocessing and faculty-reviewed prompts that generate CATME feedback through Ollama for 100+ first-year engineering students, within university data-handling policy
- Evaluate Llama, Qwen, and Kimi on VT compute, guiding faculty to models meeting performance and cost goals
- Containerize the feedback pipeline with Docker, standardizing dependencies and setup for a 100+ student cohort

Cybersecurity Intern May 2026 – Aug. 2026
Triple Point Security *Blacksburg, VA*

- Ran authenticated/unauthenticated Nessus scans on 5 hosts, surfacing 20+ critical and 84 high-severity findings
- Compared self-hosted Llama 3 with cloud LLMs for vulnerability-report generation, exposing severity errors, missed CVEs, and RAG limits that led to updated detection rules with fewer false positives
- Built an AWS–Azure site-to-site VPN, fixing an IKE mismatch; provisioned AWS, Azure, and GCP via Terraform
- Reconciled NIST and CMMC across 4 publications, defining controls for Security Hub compliance mapping

Research Assistant, Full-Stack Development Sep. 2025 – Present
Hume Center, National Security Institute & Dept. of Aerospace and Ocean Engineering *Blacksburg, VA*

- Lead development of SpaceNet, a satellite-constellation simulation testbed, building Flask REST APIs that let 5 aerospace researchers configure, launch, and monitor experiments
- Built 4 React dashboards for experiment configuration and live telemetry, used by 10+ team members
- Dockerized the platform and automated CI/CD with GitHub Actions, cutting deploys from 45 to 8 minutes

Research Assistant, Backend Development Jun. 2025 – Aug. 2025
Virginia Tech, Dept. of Computer Science (OpenDSA) *Blacksburg, VA*

- Refactored 3 Rails controllers in OpenDSA, eliminating N+1 queries and cutting database query time by 30%
- Added automated grading and real-time feedback to CS 3114's learning system in Rails, reducing manual grading

Projects

CloudGuard | *Terraform, AWS, GuardDuty, Security Hub, GitHub Actions, tfsec* Jan. 2026 – Apr. 2026

- Provisioned GuardDuty, Inspector, and Security Hub via Terraform; auto-mapped findings to NIST 800-53
- Integrated tfsec into CI to block merges that introduce infrastructure misconfigurations

WhisperKey | *Python, faster-whisper, Ollama* github.com/YoshwanPathipati/whisperkey

- Open-source, fully offline Windows dictation: hold a hotkey, speak, and formatted text lands at the cursor
- GPU speech-to-text (faster-whisper, CUDA) and a local Ollama LLM cleanup pass; no audio leaves the machine

Multimodal Associative Memory | *JAX, Modern Hopfield Networks* Feb. 2026 – Present

- Audiovisual memory with continuous-time JAX dynamics and a Modern Hopfield core for bidirectional recall
- Trained with Equilibrium Propagation's local, biologically inspired weight updates instead of backpropagation

Technical Skills

Languages: Python, Java, C, JavaScript, TypeScript, SQL, Ruby, Bash, HTML/CSS
AI/ML: JAX, Ollama, Llama 3, RAG, faster-whisper
Cloud & DevOps: AWS, Azure, GCP, Terraform, Docker, GitHub Actions, Linux, VMware
Security: Nessus, OpenVAS, Wireshark, nmap, tfsec, NIST SP 800-53 / RMF, CMMC
Web & Data: React, Node.js, Express, Flask, Django, Ruby on Rails, PostgreSQL, MySQL, MongoDB, Redis
Certifications: AWS Certified Solutions Architect – Associate (2026); CompTIA Security+ (in progress)